X-RAY generator (X-ray production)

For medical imaging, x-rays are generated by an x-ray tube. In this device, a metal filament is heated (much like a light bulb) until energetic electrons escape from the cathode surface into a vacuum. These electrons are then accelerated by an electric field, acquiring kinetic energy while being attracted to a positive anode target. The total amount of energy acquired by the electron in the accelerating electric field is equal to the product of the potential (peak kilovoltage, kVp) times the unit of electrical charge, possessing units of electron volts (kilo electron volts, keV). The amount of charge generated by the x-ray tube per unit time has units of electrical current (milli-amperes, mA), and the product of voltage and current is the amount of power (watts) delivered by the tube. Electrostatic and/or magnetic fields are used to focus the electron beam into a small area of the anode target. Typically, this focal spot has dimensions of about 1 mm. When the electrons collide with the target, most of their energy is dissipated into heat but a small fraction (<1%) is converted into several forms of electromagnetic radiation.

X-rays are released if the electrons in motion interact with matter. In the x-ray tube electrons interact with a target and a part of kinetic energy of the electrons are converted into x rays or electromagnetic energy. The potential gap of 20-150kV is generated between anode and cathode of x-ray tube by the x-ray machine. A current through a filament at the cathode side is generated by a separate low voltage circuit. The thermionic emission effect, which is the current in the filament, causes the filament to heat up and expel the electrons. The considerable potential gap among anode and cathode produces electron. The movement of electrons between anode and cathode is considered as tube voltage and the energy of electron in the cathode filament is called voltage of the filament. Bremsstrahlung process and characteristic x-ray production are the two ways of transforming the energetic electrons to x-rays at the anode side. X-rays escapes

from the tubes in both dimensions but constrained by lead boxes and collimators to the correct beam size and interact finally with the individual and the sensor for a realistic image.

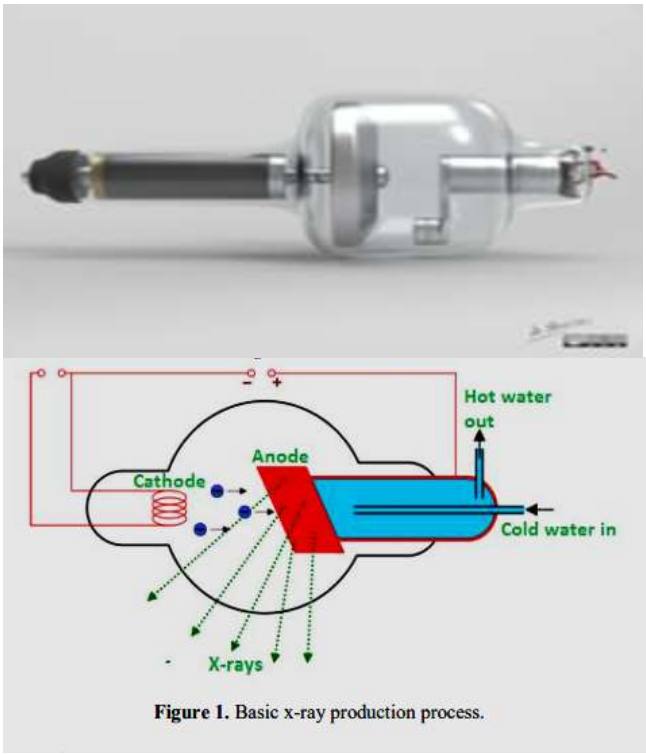


Figure 1. Basic x-ray production process.

The x-ray tube contains two principal elements:

1. filament (also acts as cathode):
   boils off electrons by thermionic emission
2. target (also acts as anode):
   electrons strike to produce x-rays

Additional components include:

- expansion bellows
  (provide space for oil to expand)
- tube envelope (evacuated)
- tube housing
- cooling dielectric oil
- rotor
- induction stator
- tube window: usually made from beryllium, not glass

X-ray production at the anode

The electrons hit the anode with a maximum kinetic energy of the kVp and interact with the anode by losing energy via:

- Elastic interaction: rare, only happens if kVp < 10 eV. Electrons interact but conserve all their energy
- Inelastic interaction: causes excitation / ionization in atoms and releases energy via electromagnetic (EM) radiation and thermal energy

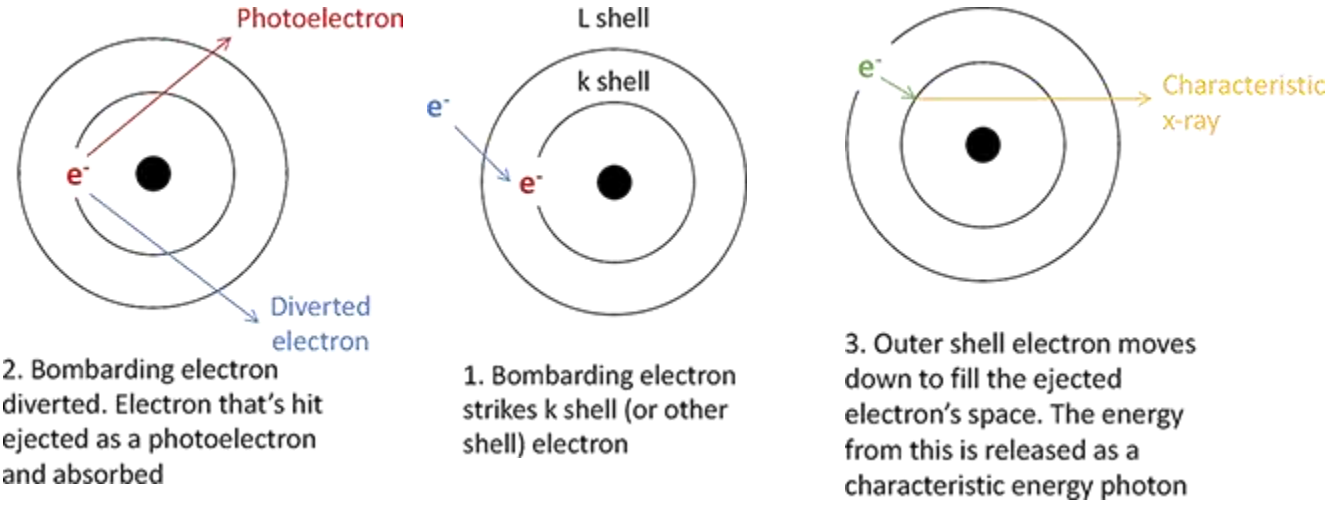
Interactions

At the anode, electrons can interact with the atoms of the anode in several ways to produce x-ray photons.

1. Outer shell interaction: low energy EM released and quickly converted into heat energy
2. Inner shell interaction: produces **characteristic radiation**
3. Nucleus field interaction: **Bremsstahlung**

1. Characteristic radiation

1. A bombarding electron knocks a k-shell or l-shell electron out.
2. A higher shell electron moves into the empty space.
3. This movement to a lower energy state releases energy in the form of an x-ray photon.
4. The bombarding electron continues on its path but is diverted.



It is called "characteristic" as energy of emitted electrons is dependent upon the anode material, not on the tube voltage. Energy is released in characteristic values corresponding to the binding energies of different shells.

For tungsten:

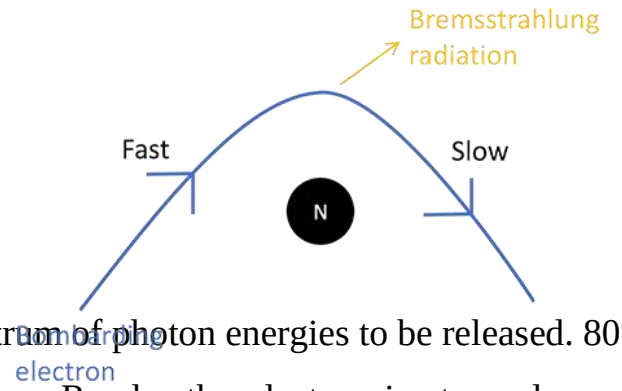
Ek – El (aka Kα) = 59.3 keV

Ek – Em (aka Kβ) = 67.6 keV

2. Bremsstrahlung

Bremsstrahlung radiation

1. Bombarding electron approaches the nucleus.
2. Electron is diverted by the electric field of the nucleus.
3. The energy loss from this diversion is released as a photon (Bremsstrahlung radiation).



Bremsstrahlung causes a spectrum of photon energies to be released. 80% of x-rays are emitted via Bremsstrahlung. Rarely, the electron is stopped completely and gives up all its energy as a photon. More commonly, a series of interactions happen in which the electron loses energy through several steps.

Cathode Assembly

The cathode unit is in the middle of electrodes for a x-ray source, a low voltage x-ray generator generates a voltage through tube filament. A focusing cup have a tungsten filament coil wire with 2- 5cm in diameter and 10-20mm in length. Filament electrons are boiled off tungsten wire. The higher voltage generates the anode-cathode electric current. The electrical field and filament arrangement form an electrical lens that determines the electron pathways and their region of influence and the region pointed to as the focal point. Cathode assembly is composed of the filament of a tungsten wire inside a focus point cup. A cathode mounting device of two filament dimensions is used for standard x-ray tubes. The big filament (right side) is used when short high intensity presentations are needed. When resolution is necessary or movement blasted (due to longer exposure), the smaller filament (left) is employed. It is desirable to have a focal point as small as

possible to fix fine detail in a radiograph. The focusing point's size is directly linked to the filament's size. The focus point size is decreased by a relatively higher heat flow charge on the destination. Thus, it is important to reduce the highest permissible power (the tube current kilovolt peak). As a consequence, the tube will also give a decrease in maximal x-ray pressure. In short exposure period, the criterion for an appropriate radiological strength defines minimum possible configuration for filament thickness. When period of exposure is too long, the picture can be movement fluttered. A cathode unit typically consists of two filaments: (1) a wider filament to accommodate high X-ray tube power applications over the short term of exposure; and (2) Smaller filament for devices with minimal resolution capacity and where the phase blurring will not be a matter of concern.

Anode Assembly

There are usually two types of anode: fixed and rotating. Originally, a tungsten reference material in a fixed copper block was used as a fixed anode. The low thermal ability reduced its strength of x-rays (i.e., radiation). The development of a revolving anode (objective) architecture allowed a significant input in x-ray intensity feasible. The instantaneous heat loaded in the middle of accelerated electrons is dispersed with a rotating anode across a much broader region. To optimize the efficiency of the split radiation, the target components utilized in an X-ray anode should be heavily radioactive, either fixed or rotating. Substantial heat charge is maintained by target material. Tungsten is used as modeling tool for x-ray source because of its high thermal conductivity, low viscosity and high atomic number. Usually to raise the thermal distortion power of rhenium by 10% tungsten is used because of its elastic and flexible nature. Usually on molybdenum surface tungsten is used even though it is not a great conductor but it has high-quality sight, and on a copper substructure as a thin pattern too.

Properties and characteristics of X-RAY:

1. In an electronic, gravitational, or hybrid field, X-rays are electrically neutral; they have no interference or variation.

2. X-rays travels straight at light speed, which is characteristic for directing and focusing rays so that they can radiate the specific area of the studied body.

3. X-rays cause biological and chemical effects that could affect the organism through the production of ionization and/or cellular changes that can lead to disorders or additional mutations.

4. X-rays occupy a part of the continuum and not only have one but many wavelengths. These depend on the number of factors leading to radiation generation.

5. The human eye or body cannot detect X-rays and thus only machines and processing procedures can be detected. In carrying out steps to secure the human body, that is an essential factor.

6. Radiographic images on photographic films, fluorescence on certain crystal types are produced by X-rays, both of which are used to produce x-rays on medically controlled monitoring films and fluoroscopes.

7. X-rays emits secondary radiation and dispersed radiation, meaning that a biological material receiving x-rays generates new rays with different characteristics in turn. These rays are typically an optical issue, and therefore harmful to the health of those operating with x-rays.